

Exercise Class 2 — Production (MWG Ch. 5)

Advanced Microeconomics 1 · Monday 19 October, 16:00–18:00

Format (120 min). 60 min: assigned students present Problems 1–3, 5 and 6 at the board (assignments posted Thursday 15 October). 30 min: instructor works the hard item (Problem 4). 30 min: timed prelim-style question (Problem 7), closed book, ungraded, self-marked against the sketch afterwards.

Also today: the two-page *course synopsis* is distributed — the map of the course plus the ten derivations you should be able to produce cold. Bring it to EC3.

Part I — Problems

Problem 1 Production set structure

- (a) Show: if Y is convex and $0 \in Y$, then Y exhibits nonincreasing returns to scale.
- (b) Show: if Y is additive and exhibits nonincreasing returns to scale, then Y is a convex cone. (Hint: build rational scalings from replication and scaling-down; use closedness for the irrational ones.)
- (c) Interpret (b) economically: what does free entry/replication do to decreasing returns at the industry level, and what does that imply for the shape of long-run industry supply?

Problem 2 The Cobb–Douglas firm, end to end

Let $f(z_1, z_2) = z_1^\alpha z_2^\beta$ with $\alpha, \beta > 0$, $\alpha + \beta < 1$; input prices $w \gg 0$, output price p .

- (a) Derive the conditional factor demands $z(w, q)$ and cost function $c(w, q)$.
- (b) Verify Shephard's lemma and the concavity of c in w (check the Hessian is NSD, or argue via the envelope-of-lines).
- (c) Derive supply $q(p, w)$ from $p = \partial c / \partial q$, then the unconditional factor demands and $\pi(p, w)$.
- (d) Verify Hotelling's lemma in both arguments: $\partial \pi / \partial p = q$ and $\partial \pi / \partial w_\ell = -z_\ell$.
- (e) What happens to each object as $\alpha + \beta \rightarrow 1$? Reconcile with the CRS pathology of the PMP.

Problem 3 The law of supply, and why no compensation

- (a) Prove: for any p, p' and profit-maximizing $y \in y(p)$, $y' \in y(p')$, $(p - p') \cdot (y - y') \geq 0$.
- (b) State the consumer-side analogue (compensated law of demand) and identify precisely which feature of the consumer's problem makes compensation necessary there and unnecessary here.
- (c) Deduce from (a): a competitive firm's output never falls when its output price rises, and its demand for an input never rises when that input's price rises. No differentiability, no convexity.

Problem 4 Wedges, dispersion, and the cost of misallocation

Two firms in the same industry share the CRS technology $Q_i = A_i L_i$ (labor the only input, wage w common), differing in productivity A_i . Firm i faces a firm-specific distortion: it behaves as if its wage were $(1 + \tau_i)w$. Output sells at a common price, normalized $P = 1$.

- (a) From profit maximization with the distortion, derive firm i 's first-order condition and show its *marginal revenue product of labor* satisfies $\text{MRPL}_i = (1 + \tau_i)w$. Conclude: absent distortions ($\tau_i = 0$ for all i), MRPLs equalize across firms *regardless of the A_i* .
- (b) An econometrician observes revenue R_i and labor L_i . Show that R_i/L_i identifies MRPL_i here, and hence that dispersion in revenue-per-worker within the industry measures dispersion in wedges, not in productivity. Why does linearity ($Q_i = A_i L_i$) matter for this step, and what changes with $Q_i = A_i L_i^\alpha$, $\alpha < 1$?
- (c) Let $A_1 = 2$, $A_2 = 1$, total labor $\bar{L} = 3$, and suppose the distortions have induced the allocation $L_1 = 1$, $L_2 = 2$. Compute aggregate output, then compute it under the allocation a planner (or undistorted prices) would choose. Report the TFP gain from reallocation in percent.
- (d) One sentence each: (i) name two real-world policies that act like a τ_i rising with firm size; (ii) explain why measurement error in R_i or L_i biases the measured gains *upward* (Bils–Klenow–Ruane 2021).

Problem 5 Gorman form: the price of the representative consumer

Consumers $i = 1, \dots, I$ have indirect utilities of the Gorman form

$$v_i(p, w_i) = a_i(p) + b(p) w_i, \quad b(p) > 0 \text{ common to all } i.$$

- (a) Apply Roy's identity to derive $x_i(p, w_i)$ and show individual demand is affine in own wealth with a wealth-slope $-\nabla_p b(p)/b(p)$ that is *identical across consumers*.
- (b) Show aggregate demand $\sum_i x_i(p, w_i)$ depends on $(w_1, \dots, w_I)$ only through $\sum_i w_i$.
- (c) Exhibit a representative consumer: an indirect utility $V(p, W)$ with $W = \sum_i w_i$ whose Roy demand equals the aggregate.
- (d) Name two familiar preference classes in Gorman form (identical homothetic; quasilinear with common numeraire), and explain in two sentences why the failure of Gorman form in the data is the reason heterogeneous-agent macroeconomics exists.

Problem 6 Efficiency and supporting prices

- (a) Prove MWG 5.F.1: if $y \in y(p)$ for some $p \gg 0$, then y is efficient.
- (b) Give an example (a picture suffices) of an efficient production plan in a *convex* Y that is profit-maximizing only for a price vector with a *zero* component; explain why $p \gg 0$ cannot be guaranteed in Prop. 5.F.2.
- (c) Give a nonconvex Y (picture) with an efficient point that is profit-maximizing for *no* price vector. Which step of the separating-hyperplane proof dies?
- (d) In two sentences: how do Hsieh–Klenow (2009) convert the *failure* of common supporting prices into a measurable object, and what is that object?

Problem 7 Timed prelim question (30 minutes, closed book)

A price-taking firm in input markets has cost function

$$c(w_1, w_2, q) = 2q\sqrt{w_1w_2}.$$

- (a) Verify that c has the properties a cost function must have: homogeneous of degree one, nondecreasing, and concave in w ; nondecreasing in q .
- (b) Recover the conditional factor demands $z(w, q)$ by Shephard's lemma, and identify the underlying technology.
- (c) The firm sells its output at price p , taking p as given. Derive the supply correspondence and the profit function. Comment on the returns to scale.
- (d) You observe: expenditure on the input bundle equal to 60, revenue equal to 90, and you are told the technology is as above (so the output elasticity of the input bundle is 1). Show the firm behaves as if its input prices carried a wedge $(1 + \tau) = 1.5$. Give *two* economic stories consistent with this number, and state in one sentence what additional data would distinguish them.